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TRANSPORT PHENOMENA I

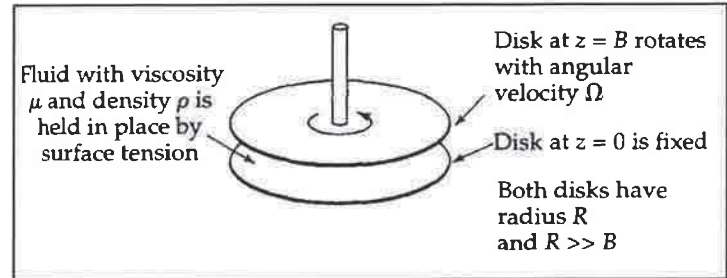
FALL 2025

EXAM 2

60
40
100

PARALLEL-DISK VISCOMETER

A fluid is placed in the gap between two horizontal circular disks of radius R . Disks are separated by a small distance B . The lower disk is held fixed. The upper disk is rotating at a constant angular velocity Ω .



Postulates:

- Newtonian fluid with constant density ρ and constant viscosity μ .
- $v_r = 0$ and $v_z = 0$.
- Pressure in the fluid is not a function of θ .
- Creeping flow assumption: The flow is very slow and the acceleration term $\frac{D\langle v \rangle}{Dt}$ in the equation of motion is negligible.

15/20 1. Using the continuity equation, prove v_θ is not a function of θ at steady-state conditions.

20/20 2. Using the proper equation of motion at steady-state conditions, write down and simplify a partial differential equation for v_θ as a function of r and z .

5/20 3. Assuming v_θ may be written as r multiplied by a function of z ,
$$v_\theta(r, z) = r f(z)$$

, convert the partial differential equation for v_θ to an ordinary differential equation for f .

15/20 4. Knowing $v_\theta(r, z = 0) = 0$ and $v_\theta(r, z = B) = r \Omega$, write down two boundary conditions for f , solve the ordinary differential equation derived in part 1.3. and evaluate the integration constants.

5/20 5. One can estimate the viscosity of the fluid by measuring the torque T_z required to rotate the upper disk at a constant angular velocity. Using the velocity profile derived in part 1.4., find an equation for the viscosity in terms of T_z, Ω, B and R . Hint: torque exerted by the liquid on unit area of the upper disk is the product of the radius r (lever arm) and viscous stress in θ -direction.

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§B.1 NEWTON'S LAW OF VISCOSITY

$$[\tau = -\mu(\nabla \mathbf{v} + (\nabla \mathbf{v})^t) + (\frac{2}{3}\mu - \kappa)(\nabla \cdot \mathbf{v})\delta]$$

Cylindrical coordinates (r, θ, z) :

$$\tau_{rr} = -\mu \left[2 \frac{\partial v_r}{\partial r} \right] + (\frac{2}{3}\mu - \kappa)(\nabla \cdot \mathbf{v}) \quad (\text{B.1-8})^a$$

$$\tau_{\theta\theta} = -\mu \left[2 \left(\frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} \right) \right] + (\frac{2}{3}\mu - \kappa)(\nabla \cdot \mathbf{v}) \quad (\text{B.1-9})^a$$

$$\tau_{zz} = -\mu \left[2 \frac{\partial v_z}{\partial z} \right] + (\frac{2}{3}\mu - \kappa)(\nabla \cdot \mathbf{v}) \quad (\text{B.1-10})^a$$

$$\tau_{r\theta} = \tau_{\theta r} = -\mu \left[r \frac{\partial}{\partial r} \left(\frac{v_\theta}{r} \right) + \frac{1}{r} \frac{\partial v_r}{\partial \theta} \right] \quad (\text{B.1-11})$$

$$\tau_{\theta z} = \tau_{z\theta} = -\mu \left[\frac{1}{r} \frac{\partial v_z}{\partial \theta} + \frac{\partial v_\theta}{\partial z} \right] \quad (\text{B.1-12})$$

$$\tau_{zr} = \tau_{rz} = -\mu \left[\frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \right] \quad (\text{B.1-13})$$

in which

$$(\nabla \cdot \mathbf{v}) = \frac{1}{r} \frac{\partial}{\partial r} (rv_r) + \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z} \quad (\text{B.1-14})$$

^a When the fluid is assumed to have constant density, the term containing $(\nabla \cdot \mathbf{v})$ may be omitted. For monatomic gases at low density, the dilatational viscosity κ is zero.

§B.4 THE EQUATION OF CONTINUITY^a

$$[\partial \rho / \partial t + (\nabla \cdot \rho \mathbf{v}) = 0]$$

Cylindrical coordinates (r, θ, z) :

$$\frac{\partial \rho}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r} (\rho r v_r) + \frac{1}{r} \frac{\partial}{\partial \theta} (\rho v_\theta) + \frac{\partial}{\partial z} (\rho v_z) = 0 \quad (\text{B.4-2})$$

§B.6 EQUATION OF MOTION FOR A NEWTONIAN FLUID WITH CONSTANT ρ AND μ

$$[\rho D\mathbf{v}/Dt = -\nabla p + \mu \nabla^2 \mathbf{v} + \rho \mathbf{g}]$$

Cylindrical coordinates (r, θ, z) :

$$\rho \left(\frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_r}{\partial \theta} + v_z \frac{\partial v_r}{\partial z} - \frac{v_\theta^2}{r} \right) = -\frac{\partial p}{\partial r} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (rv_r) \right) + \frac{1}{r^2} \frac{\partial^2 v_r}{\partial \theta^2} + \frac{\partial^2 v_r}{\partial z^2} - \frac{2}{r^2} \frac{\partial v_\theta}{\partial \theta} \right] + \rho g_r \quad (\text{B.6-4})$$

$$\rho \left(\frac{\partial v_\theta}{\partial t} + v_r \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} + v_z \frac{\partial v_\theta}{\partial z} + \frac{v_r v_\theta}{r} \right) = -\frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (rv_\theta) \right) + \frac{1}{r^2} \frac{\partial^2 v_\theta}{\partial \theta^2} + \frac{\partial^2 v_\theta}{\partial z^2} + \frac{2}{r^2} \frac{\partial v_r}{\partial \theta} \right] + \rho g_\theta \quad (\text{B.6-5})$$

$$\rho \left(\frac{\partial v_z}{\partial t} + v_r \frac{\partial v_z}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_z}{\partial \theta} + v_z \frac{\partial v_z}{\partial z} \right) = -\frac{\partial p}{\partial z} + \mu \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_z}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 v_z}{\partial \theta^2} + \frac{\partial^2 v_z}{\partial z^2} \right] + \rho g_z \quad (\text{B.6-6})$$

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$$v_r = v_z = 0$$

$$v_\theta(\theta) = 0$$

$$v_\theta(r, z) \neq 0$$

1) using (B.4-2) :-

$$\cancel{\frac{\partial v_\theta}{\partial t}} + \frac{1}{r} \frac{\partial}{\partial r} (r v_r v_\theta) + \frac{1}{r} \frac{\partial}{\partial \theta} (r v_\theta v_\theta) + \frac{\partial}{\partial z} (v_z v_\theta) = 0$$

steady state
 $v_r = 0$
 $v_\theta \neq v_\theta(\theta)$
 $v_z = 0$

$$0 = 0$$

2) using (B.6-5) $v_\theta(r, z)$

$$\cancel{\rho \left(\frac{\partial v_\theta}{\partial t} + v_r \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} + v_z \frac{\partial v_\theta}{\partial z} \right)} + \cancel{\mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) + \frac{1}{r^2} \frac{\partial^2 v_\theta}{\partial \theta^2} + \frac{\partial^2 v_\theta}{\partial z^2} + \frac{2}{r^2} \frac{\partial v_r}{\partial \theta} \right]} = \cancel{\rho g_\theta}$$

steady state
 $v_r = 0$
 $v_\theta = v_\theta(r, z)$
 $v_z = 0$
 $\frac{\partial v_\theta}{\partial \theta} = 0$
 $v_\theta(\theta) = 0$
 $v_r = 0$
 we do have gravity pointing downwards $-g \sin \theta$

The equation becomes:-

$$0 = \mu \left(\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) + \frac{\partial^2 v_\theta}{\partial z^2} \right)$$

$\mu \neq 0$

$$\int \left(\frac{\partial^2 v_\theta}{\partial r^2} + \frac{\partial^2 v_\theta}{\partial z^2} \right) = 0$$

$$\frac{v_\theta}{\partial r} + \frac{v_\theta}{\partial z} = C_1$$

$$v_\theta + v_\theta = C_1 r z + C_2$$

$$v_\theta = \frac{C_1 r z}{2} + \frac{C_2}{2} \leftarrow \text{simplified}$$

integrating twice and adding the integration constants; we have to integrate with respect to both r and z .

3) $v_\theta(r, z) = r f(z)$ $\frac{v_\theta}{z} = C_1 \rightarrow v_\theta = C_1 z + C_2$

$$f(z) = \frac{v_\theta(z)}{r} = C_1 z + C_2$$

4) $v_\theta(r, z=0) = 0$ B.C.1
 $v_\theta(r, z=B) = r\Omega$ B.C.2

The new boundary conditions:-

$$f(z) = \frac{v_\theta(r, z=0)}{r} = 0$$

$$f(z) = \frac{v_\theta(r, z=B)}{r} = \frac{r\Omega}{r} = \Omega$$

changing the boundary conditions:-

$$f(z) = 0$$

$$f(z=B) = \Omega$$

Boundary conditions

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4)

$$\left. \begin{aligned} f(z) &= c_1 z + c_2 \\ f(z=0) &= 0 \\ f(z=B) &= \Omega \end{aligned} \right\} \begin{array}{l} \text{using this equation} \\ \text{and these boundary} \\ \text{conditions} \end{array}$$

Solving for c_1 and c_2

$$\left. \begin{aligned} 0 &= c_1 z + c_2 \\ c_1 &= \frac{c_2}{z} \end{aligned} \right\} \begin{array}{l} \text{From} \\ \text{B.C.1} \end{array}$$

$$\Omega = c_1 z + c_2 \rightarrow \text{From B.C.2}$$

$$\Omega = c_2 + c_2 \rightarrow \boxed{c_2 = \frac{\Omega}{2}}$$

and

$$\boxed{c_1 = \frac{\frac{\Omega}{2}}{z} = \frac{\Omega}{2z}}$$

5) Solving for the viscous stress

$$\tau_{\theta z} = -\mu \frac{\partial v_{\theta}}{\partial z} = -\mu r z \left(c_1 z + c_2 \right) = -\mu r z \left(\frac{\Omega}{2z} + \frac{\Omega}{2} \right)$$

$$= -\mu r z \Omega \left(\frac{1}{2z} + \frac{1}{2} \right)$$

$$v_{\theta} = r \times f(z)$$

$$@ z=0 \quad \tau_{\theta z} = 0$$

$$@ z=B \quad \tau_{\theta z} = -\mu r B \Omega \left(\frac{1}{2B} + \frac{1}{2} \right)$$

and finally at $z=B$ and $r=R$.

$$\boxed{T_z = -\mu R B \Omega \left(\frac{1}{2B} + \frac{1}{2} \right)}$$

we're looking

at viscous stress

in z -direction and also rotating

so I will use $\tau_{\theta z} = \tau_{z\theta}$

Newton's law of viscosity Equation B.1-12

$$\tau_{z\theta} = \tau_{\theta z} = -\mu \left[\frac{1}{r} \frac{\partial v_z}{\partial \theta} + \frac{\partial v_{\theta}}{\partial z} \right]$$

it's not a function of theta and we're looking at it as a function of z .

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5) using Newton's law of viscosity:

using equation 7.02 (B.1-12)

because we're looking at the viscous stress in the z -direction
and our disk is rotating in the θ -direction:

Equation:

$$\tau_{z\theta} = \tau_{\theta z} = -\mu \left[\frac{1}{r} \frac{\partial v_{\theta}}{\partial \theta} + \frac{\partial v_{\theta}}{\partial z} \right]$$

$\Rightarrow -\mu \frac{\partial v_{\theta}}{\partial z}$

$v_{\theta} = 0$

This is my first
I believe it's
incorrect

continuing part 4:-

$$f(z) = \frac{C_1 z}{2} + \frac{C_2}{2r}$$

and the boundary conditions

$$f(r=R, z=0) = 0$$

$$f(r=R, z=b) = R$$

Solving for the integration constants:-

$$0 = \frac{C_1 z}{2} + \frac{C_2}{2r} \rightarrow \text{from B.C. 1}$$

$$C_1 = -\frac{C_2}{2r} \times \frac{2}{z} = -\frac{C_2}{rz}$$

$$R = \frac{-\frac{C_2}{rz} \times z}{2} + \frac{C_2}{2r} \rightarrow \text{from B.C. 2}$$

$$R = -\frac{C_2}{2r} + \frac{C_2}{2r} = 0$$

Questionable!

Looking at B.C. 2:-

$$R = \frac{C_1 z}{2} + \frac{C_2}{2r}$$

$$C_1 = \left(R - \frac{C_2}{2r} \right) \frac{2}{z}$$

$$0 = \frac{\left(R - \frac{C_2}{2r} \right) \frac{2}{z} \times z}{2} + \frac{C_2}{2r}$$

$0 = R$ again!

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